Artificial Intelligence
(Lesson 1)

AI IN FICTION & ASIMOV'S LAWS

Famous AI Fiction:
◆ Frankenstein (1818) – created being vs creator (AI ethics precursor)
◆ 2001: A Space Odyssey (1968) – HAL 9000, sinister AI ◆ Galatea myth (ancient)

Asimov's 3 Laws of Robotics:
1. Cannot harm humans 2. Must obey humans 3. Protect its own existence

These laws anticipate modern AI alignment & safety research challenges.

WHAT IS AI?

Definition:
Ability of a machine to think & act like a human being.

Key AI Branches:
◆ Machine Learning
◆ Natural Language Processing (NLP)
◆ Neural Networks & Deep Learning
◆ Robotics ◆ Generative AI (LLMs)
◆ Computer Vision ◆ Expert Systems

Examples:
ChatGPT, Gmail Smart Reply, Amazon Recommendations, Face ID

IBM: What is Artificial Intelligence?

AI FIELDS & EXAMPLES

Fields:
ess AI, NPCs, pathfinding)
2. Fuzzy Logic (washing machines, cameras)
3. Genetic Algorithms (optimisation)
4. Expert Systems (medical diagnosis)
5. Data Mining (Amazon, Netflix recs)
6. Neural Networks (face & speech recog)
7. Robotics (Boston Dynamics, SOPHIA)
8. Natural Language Processing
(Siri, Alexa, Google Translate)

My observation:
Every major tech product today incorporates at least one AI field.

MIT OpenCourseWare – AI Course

HISTORY

Founders (1956 Dartmouth):
◆ John McCarthy
coined "Artificial Intelligence"
◆ Marvin Minsky
MIT AI Lab co-founder
◆ Alan Turing (1950)
"Computing Machinery & Intelligence" paper

Milestones:
1950→Turing Test 1956→AI coined
1997→Deep Blue 2022→ChatGPT

BBC: Alan Turing

AI TRAITS

Intelligence requires:
◆ Decision Making
◆ Perception (see, hear, feel)
◆ Judgement ability
◆ Searching / Pathfinding
◆ Learning & Remembering
◆ Natural Language
◆ Prediction
◆ Creativity

"Creativity is the hardest trait to replicate in AI"

Stanford Encyclopedia – AI

SYNTHESIS

Turing Test ↔ Chinese Room
Strong AI ↔ Weak AI ↔ AGI

TURING TEST (1950)

Alan Turing's "Imitation Game":
If a human cannot distinguish the machine from a human → machine passes the test.

Supports AI Intelligence ✓
Modern LLMs (ChatGPT, Claude) routinely pass conversational Turing tests.

My view: Passing ≠ understanding. Behaviour is not the same as consciousness.

Turing's 1950 Paper – Mind Journal

CHINESE ROOM (Searle)

John Searle (1980) – Against AI:
A person follows symbol rules without understanding meaning → like a computer.

Conclusion:
Computers can NEVER have true intelligence. No consciousness. No self-awareness.

Counters Turing Test — syntax ≠ semantics.

Stanford: Chinese Room Argument

STRONG vs WEAK AI

Strong AI (General AI):
Human-level intelligence across all domains.
e.g. SOPHIA robot, ChatGPT (near-general)
Millions of lines of code, years to develop.

Weak AI (Narrow AI):
Task-specific, limited domain only.
e.g. Spam filters, Chess AI, Chatbots
Small codebase, quick to build.

Hanson Robotics – SOPHIA

FUTURE

◆ IoT – Smart devices
◆ Smart Homes
◆ AGI on the horizon
◆ Robots with rights?
◆ *Existential risk?*